L&T ASSIGNMENT REPORT

AUTOMATIC NIGHT LAMP CONTROLLER PCB

PCB Design using KiCad

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Course / Section	ELCE Sec: A
Assignment Title	Automatic Night Lamp Controller PCB
Design Software	KiCad EDA

Abstract

This project presents the design of a simple automatic night lamp controller. An LDR senses the surrounding light level and a BC547B NPN transistor acts as an electronic switch. The circuit turns the LED on when the surroundings become dark and switches it off when daylight is present. KiCad is used to document the schematic, assign component footprints and design the PCB layout.

Project objectives

- Detect changes in ambient light using an LDR.
- Switch an LED automatically using a transistor.
- Create a clear schematic and assign practical footprints in KiCad.
- Review the PCB in 2D and 3D views.

1. Theory and Working Principle

An automatic night lamp is a light-sensitive electronic circuit that controls a lamp without manual switching. It is useful for pathways, staircases, gardens and home lighting where a lamp should operate only when the surroundings are dark.

Light Dependent Resistor (LDR)

An LDR is a photoresistive component whose resistance changes with light intensity. Its resistance is low in bright light and increases in darkness. The LDR forms part of a sensing network that changes the voltage applied to the transistor's base.

NPN transistor as a switch

The BC547B is an NPN bipolar junction transistor. A small base current controls the larger collector current. When the base-emitter junction is sufficiently forward biased, the transistor turns on and allows current to flow through the LED branch. When base current is insufficient, the transistor turns off.

LED and current-limiting resistor

The LED produces light when forward biased. A series resistor limits its current and protects the LED and transistor. The resistor value depends on the supply voltage, LED forward voltage and desired current.

Circuit operation

In daylight: the LDR resistance falls, the sensing node is pulled toward the low-voltage side, and the transistor switches off; the LED is off.
In darkness: the LDR resistance rises, the sensing node voltage increases through the bias resistor, the transistor switches on, and current flows through the LED; the LED lights.

Design note: Verify the exact resistor values and the transistor pinout against the actual schematic and component datasheets before powering or fabricating the board.

2. KiCad Schematic

The schematic documents the circuit symbols, component values, power connections and electrical connections.

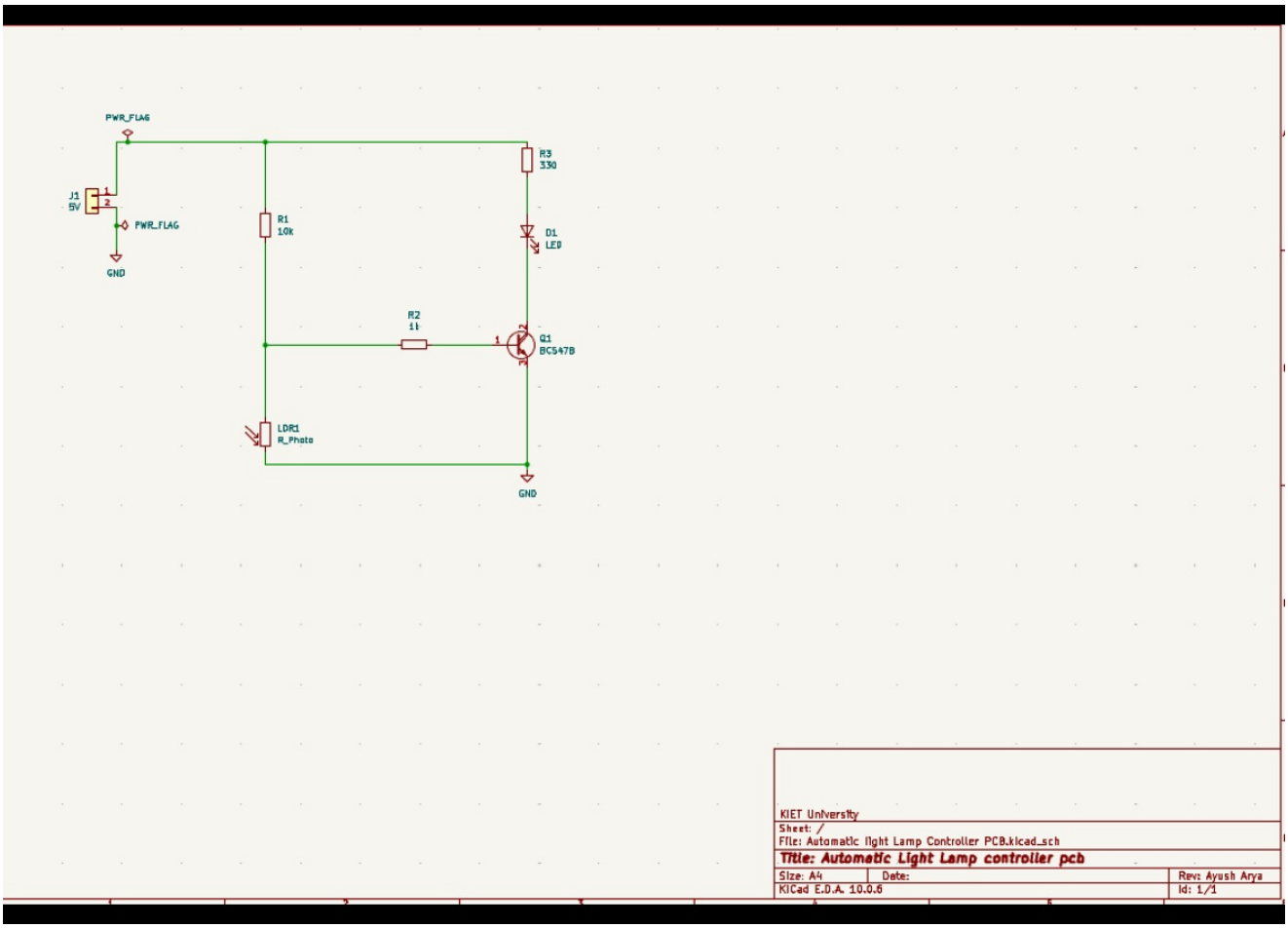
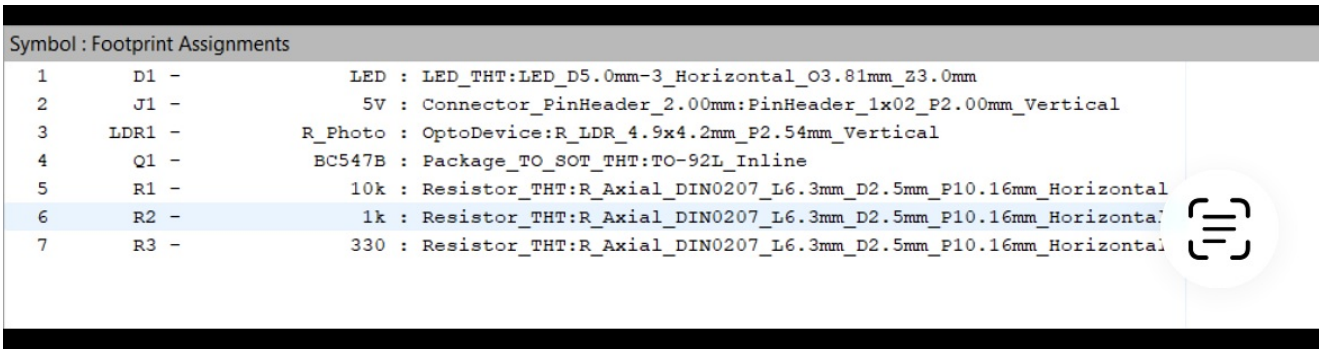


Figure 1. Actual schematic screenshot supplied by the student.

3. Footprint Assignment

Footprints map each schematic symbol to the physical pad layout required for PCB assembly.



Symbol : Footprint Assignments

1	D1 -	LED	LED_THT:LED_D5.0mm-3_Horizontal_O3.81mm_Z3.0mm
2	J1 -	5V	Connector_PinHeader_2.00mm:PinHeader_1x02_P2.00mm_Vertical
3	LDR1 -	R_Photo	OptoDevice:R_LDR_4.9x4.2mm_P2.54mm_Vertical
4	Q1 -	BC547B	Package_TO_SOT_THT:TO-92L_Inline
5	R1 -	10k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
6	R2 -	1k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
7	R3 -	330	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal

Figure 2. Actual KiCad footprint assignment screenshot supplied by the student.

4. 2D PCB View

The 2D PCB Editor view is used to position footprints, inspect board geometry and route copper tracks between pads.

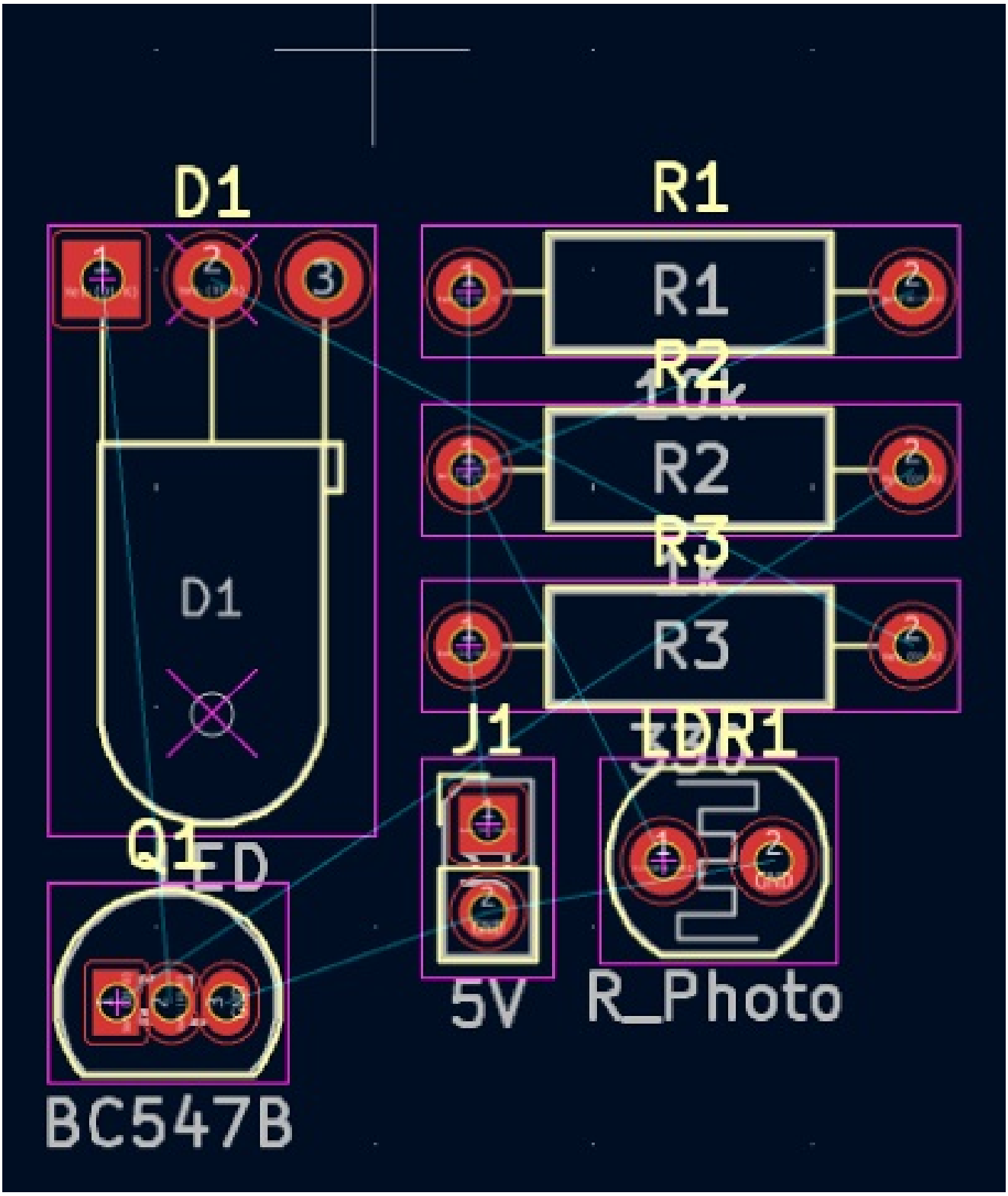


Figure 3. Actual 2D PCB screenshot supplied by the student.

5. 3D PCB View

The 3D view helps visually inspect component placement, orientation and the overall board appearance.

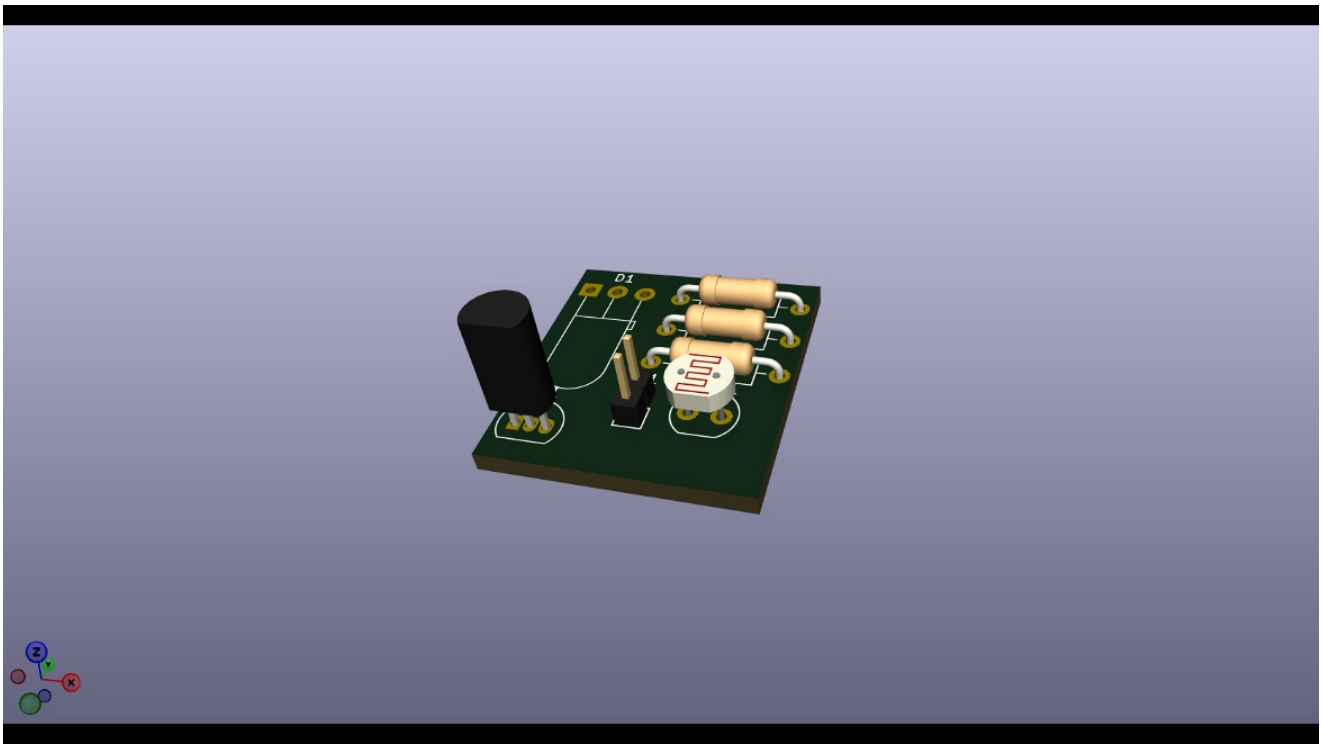


Figure 4. Actual KiCad 3D PCB screenshot supplied by the student.

6. Bill of Materials (BOM)

The following BOM is based on the component references and values visible in the supplied screenshots. Confirm it against the final KiCad schematic before ordering parts.

Item	Reference	Description / Value	Qty	Footprint / Notes
1	J1	5 V power connector / 2-pin header	1	2.00 mm pin header as shown
2	LDR1	Photoresistor (LDR)	1	R_Photo footprint
3	Q1	BC547B NPN transistor	1	TO-92 inline
4	D1	LED	1	5 mm horizontal LED
5	R1	10 kΩ resistor	1	Axial through-hole
6	R2	1 kΩ resistor	1	Axial through-hole
7	R3	330 Ω resistor	1	Axial through-hole

7. PCB Design Considerations

• Place the LDR where it can receive ambient light without being blocked by tall components.
• Keep the LED and its series resistor close together.
• Use suitable track widths and clearances for the fabrication process.
• Check the transistor pin mapping and connector polarity.
• Run Electrical Rules Checker (ERC) on the schematic and Design Rules Checker (DRC) on the PCB before fabrication.

Safety: This is a low-voltage DC circuit. Do not connect it directly to mains voltage.

8. Conclusion

The Automatic Night Lamp Controller demonstrates how an LDR, transistor, resistors and LED can be combined to automate lighting based on ambient brightness. The KiCad screenshots document the schematic, footprint assignment, 2D PCB layout and 3D view. The final board should be electrically checked and compared with the physical component dimensions before fabrication.

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